§12-2 传热学一般公式

(纯导热)

$$\Phi = \lambda A \frac{t_1 - t_2}{\delta}$$

$$q = \frac{\Phi}{A} = \frac{\lambda}{\delta}(t_1 - t_2)$$

对比 $q = k(t_H - t_L)$

$$k = \frac{\lambda}{\delta}$$

(纯对流)

$$\Phi = hA(t_w - t_f)$$

$$q = \frac{\Phi}{A} = h(t_w - t_f)$$

$\therefore k = h$ (经对比)

(纯辐射)

$$q = \varepsilon\sigma(T_1^4 - T_2^4)$$

引入 $q = h_r(T_1 - T_2)$

其中 $k = h_r = \dfrac{\varepsilon\sigma(T_1^4 - T_2^4)}{T_1 - T_2}$

$\therefore$ (复合传热下的 k)

① (对流 + 导热 + 对流)

$$R_{total} = R_{h_1} + R_\lambda + R_{h_2}$$

$R_{h_1} = \frac{1}{h_1}$, $R_{h_2} = \frac{1}{h_2}$, $R_\lambda = \frac{\delta}{\lambda}$

$$k_{串} = \frac{1}{\frac{1}{h_1} + \frac{\delta}{\lambda} + \frac{1}{h_2}}$$

② (对流 + 辐射)

$q = q_{对流} + q_{辐射}$

$q = h(\quad) + h_r(\quad)$

$$k_{并} = h + h_r$$

~~串联~~ $k = k_1 + k_2$ 并

~~串联~~ $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$ 串

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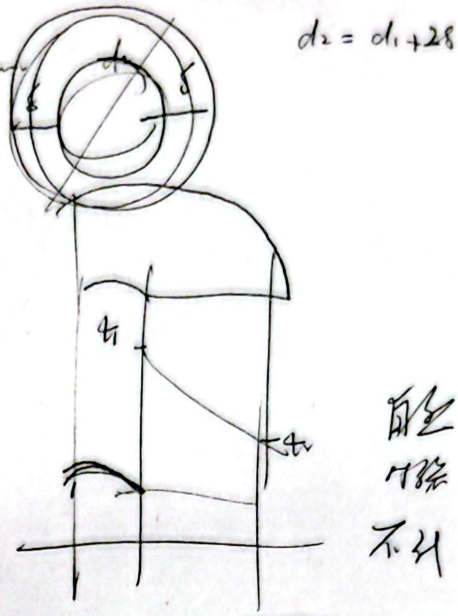
Q 12-3 $\quad t_f = 80℃$, $u_\infty = 5\,m/s$, $d_1 = 80mm$, $\delta = 10mm$, $d_2 = 160mm$, $d_2 = d_1 + 2\delta$

$\lambda = 45\,W/mK$, $t_\infty = 20℃$



$q_l = \frac{q}{l} \Rightarrow l = 1m$,

$\cancel{R_{总} = \frac{1}{\frac{1}{h_1} + R_{wall} + \frac{1}{h_2}}}$, R_{wall}

$R_{total,l} = R_{h_1,l} + R_{\lambda,l} + R_{h_2,l}$

管内对流

$R_{h_1,l} = \frac{1}{h_1 2\pi r_1} = \frac{1}{h_1 \pi d_1}$, $\quad R_{\lambda,l} = \frac{1}{2\pi\lambda} \ln\frac{d_2}{d_1}$

$\because A = 2\pi r l$

$= \frac{1}{2785(\pi)(0.08)} \quad = \frac{1}{2\pi(45)} \ln\left(\frac{0.1}{0.08}\right)$

$\approx 1.42 \times 10^{-3} \quad \approx 7.72 \times 10^{-4}$

D-B 公式

$Nu = 0.023 Re^{0.8} Pr^{0.4}$,

$h = \frac{Nu\lambda}{d}$

查水的物性,

$Nu = 330$, $h \approx 2785\,W/m^2K$

$\cancel{R_l = 2.192 \times 10^{-3}}$

$R_{h_2,l} = \frac{1}{h_2 \pi d_2}$, $\quad h_2 = \frac{Nu\lambda_2}{d_2}$

$Nu = 0.53\, Gr^{0.25} Pr^{0.25}$

(膜温) $\quad t_m = \frac{t_{w2} + t_{f2}}{2} \approx 45℃$, 假设 $\left\{ \boxed{t_{w} = 75℃},\ t_{f2} = 20℃ \right.$

$Gr = \frac{g\beta(t_{w2} - t_{f2})d_2^3}{\nu^2} \approx 5 \times 10^6$

$Pr = \frac{c_p\mu}{\lambda_2} = 0.70$, $\quad \therefore Nu \approx 195$, $h_2 \approx 5.46\,W/m^2K$

$R_{h_2,l} = \frac{1}{h_2 \pi d_2} \approx 0.580$

$R_{total,l} = 0.582\ (mK)/W$

$\Phi_l = \frac{t_{f1} - t_{f2}}{R_{total,l}} = \frac{80-20}{0.582} \approx 103\,W/m$

$\Phi_l = h_2 \pi d_2 (t_{w} - t_{f2})$

$t_w = t_f + \frac{\Phi_l}{h_2 \pi d_2} = 20 + \frac{103}{5.46 \times \pi \times 0.1} \approx 78.5℃$

(迭代法)